CSE621 - Spring 2026

Project 3: Link and Social Network Analysis

Gephi-based analysis of the Zachary Karate Club benchmark network

Team Members	Yugal Kanukolanu; Varshika Kotapati
Project and Date	Project 3 | March 22, 2026
Tool Used	Gephi
Video Link	SharePoint video: Open video link https://cardmaillouisville-my.sharepoint.com/:v:/g/personal/v0kanu01_louisville_edu/IQDjb5TOLWcvTon62tdfGk-gAVMV3nQkouqEdIXItYuwT4c
Code Link	GitHub repository: Open code repository https://github.com/kvssyugal/cse621-project3-sna

1. Objective and Dataset

The purpose of this project is to examine how a social graph reveals structure, connectivity, and community formation when it is analyzed visually and quantitatively. The benchmark used in this report is Zachary's Karate Club, a classic undirected social network containing 34 nodes and 78 edges.

Gephi was used to generate the layout, compute core network statistics, and run modularity-based community detection at multiple resolution settings. Because this network is small enough to inspect directly but still rich enough to show hubs, bridges, and factional structure, it is a strong benchmark for this assignment.

Dataset	Zachary Karate Club
Source	Gephi desktop documentation benchmark dataset
Graph type	Undirected, unweighted
Nodes	34
Edges	78
Why this dataset matters	It is widely used because the club historically split into factions, making it useful for community detection and centrality analysis.

2. Visualization Workflow in Gephi

The network was imported into Gephi and arranged with a ForceAtlas2 layout. This layout is useful because it spreads the graph in a way that makes hubs and dense neighborhoods easier to see. The first view below is the basic force-directed layout before community coloring.

Even in the plain black-and-white view, the graph already suggests a strong split between a left-side cluster and a denser right-side region connected through a smaller set of bridging links.

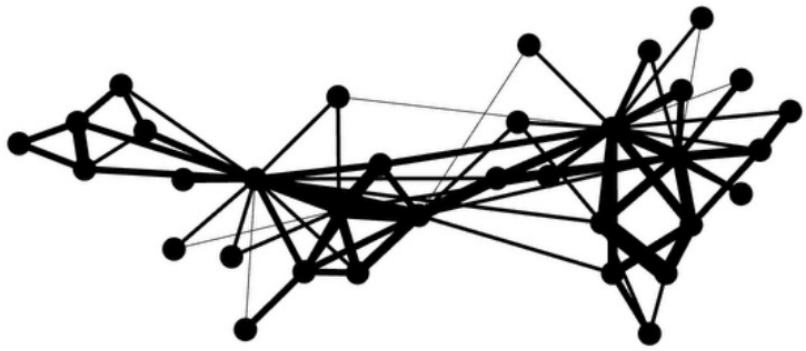
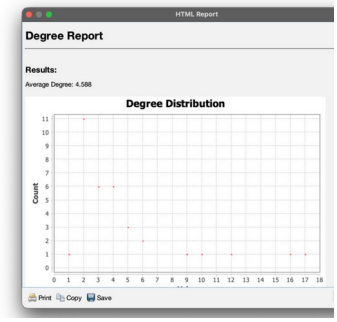


Figure 1. ForceAtlas2 layout generated in Gephi

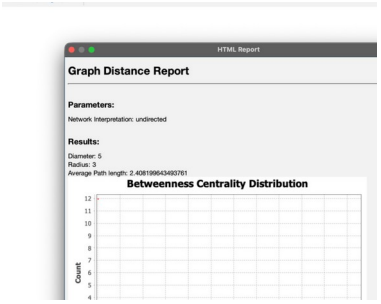
3. Key Network Metrics

Gephi produced the core network statistics shown below. The values are consistent with the expected structure of the Zachary Karate Club graph: a relatively sparse network with short path lengths and a small set of highly connected nodes.

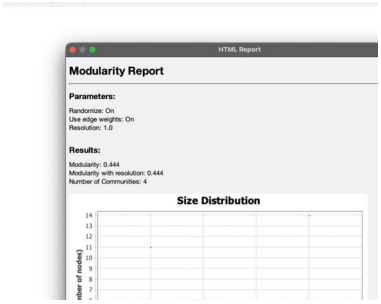
Metric	Value / Interpretation
Average degree	4.588. On average, each node is connected to a little more than four other nodes.
Diameter	5. The longest shortest-path distance between any two nodes is five steps.
Radius	3. The most central nodes are at most three steps away from all others.
Average path length	2.408. Even though the graph is sparse overall, most nodes can still be reached quickly.
Graph density	0.139 based on 34 nodes and 78 undirected edges, showing that only a fraction of all possible ties are present.



Degree report



Graph distance report



Modularity report (resolution = 1.0)

4. Community Detection Results

Community detection was run with Gephi's modularity tool. The first run used the default resolution of 1.0 and produced 4 communities with modularity 0.444. The class shares shown in Gephi correspond to approximate community sizes of 14, 11, 5, and 4 nodes.

The second run used resolution 1.5. In this Gephi run the partition simplified to 2 larger communities with modularity 0.404. That result is important because it shows how sensitive community structure can be to parameter choices. Instead of assuming that a higher setting always gives finer groups, the analyst has to examine the actual output.

Run	Resolution	Communities	Modularity
A	1.0	4 communities (about 14, 11, 5, 4 nodes)	0.444
B	1.5	2 communities (about 18 and 16 nodes)	0.404

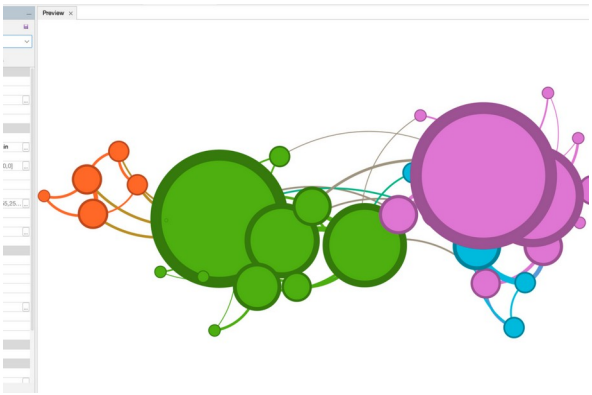


Figure 2. Community view at resolution = 1.0 (4 communities)

Figure 3. Community view at resolution = 1.5 (2 communities)

5. Analysis and Discussion

The visualization and metric results tell a consistent story. The network is not uniformly connected. Instead, it is organized around a few large hub-like nodes that anchor the main neighborhoods. This is why the average degree is moderate while the average path length remains low: a handful of well-connected nodes keep the graph compact.

The diameter of 5 and radius of 3 also support that interpretation. No pair of nodes is extremely far apart, which means the network has a small-world feel even though the graph is not dense. In practical terms, information or influence can move across the club fairly quickly before the split becomes visible in the community structure.

The community comparison is especially important. At resolution 1.0, Gephi separated the graph into 4 groups, which exposed smaller sub-communities inside the broader camps. At resolution 1.5, the graph collapsed into 2 larger communities with slightly lower modularity. This contrast shows that the reported partition depends on the resolution setting and that the analyst should justify the chosen setting rather than reporting only one run without comparison.

6. Conclusion

Using Gephi, this project successfully visualized the Zachary Karate Club network, measured core graph properties, and compared community structure under two modularity settings. The final results show a sparse but compact social network with short paths, a small number of dominant hubs, and community structure that changes visibly when the resolution parameter is adjusted. Overall, the project demonstrates why visualization, descriptive metrics, and community detection should be interpreted together rather than in isolation.

7. References

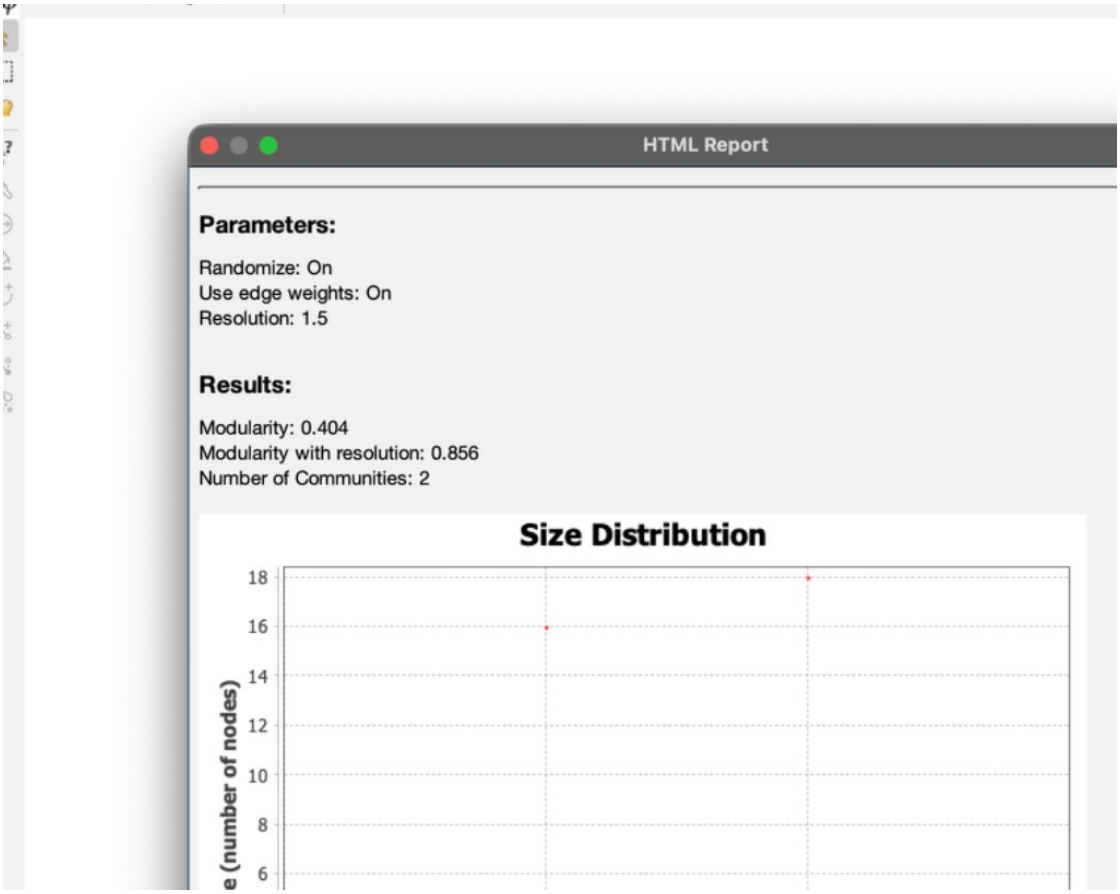
Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. Journal of Statistical Mechanics: Theory and Experiment, P10008.

Gephi Desktop Documentation. Datasets and user manual materials for benchmark graphs and community detection workflows.

Zachary, W. W. (1977). An information flow model for conflict and fission in small groups. Journal of Anthropological Research, 33(4), 452-473.

Appendix: Additional Gephi Evidence

The screenshot below records the Gephi modularity report for the second run at resolution 1.5, where the graph split into 2 communities and the reported modularity value was 0.404.



Appendix Figure A1. Gephi modularity report at resolution = 1.5